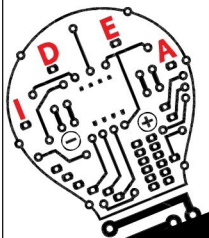


Original, innovative circuit ideas are invited

from designers and
experimenters
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PARTS LIST

Semiconductors:	
IC1, IC2, IC3	- CD4029N up/down counter
IC4-IC14	- 74LS173N 4-bit D-type register with tristate outputs
IC15, IC16	- 74LS157N quad 2-line to 1-line multiplexer
IC17	- CY6264 8k × 8 static RAM
IC18, IC19	- AT28C64 EEPROM
IC20-IC24, IC29-IC42, IC47	- 74LS245N tristate octal bus transceiver
IC25, IC26	- 74LS04N hex inverter
IC27, IC28, IC46	- 74LS08N quad 2-input AND gate
IC29, IC30	- 74LS32N quad 2-input OR gate
IC31-IC34	- 74LS86N quad 2-input exclusive OR gate
IC35, IC36	- 74LS85N 4-bit magnitude comparator
IC37, IC38	- 74LS83N 4-bit binary full adder with fast carry
IC43, IC44	- 74LS42N BCD to decimal decoder
IC45	- NE555 timer
Resistors (all 1/4-watt, ±5% carbon):	
R1-R8, R13-R20	- 1-kilo-ohm
R9-R12, R21-R28	- 100-ohm
VR1	- 100-kilo-ohm potentiometer
Capacitors:	
C1	- 47µF, 16V electrolytic
C2, C4, C5, C7, C8	- 0.01µF ceramic
C3	- 1µF, 16V electrolytic
C6	- 10µF, 16V electrolytic
Miscellaneous:	
S1, S2	- 8-pin DIP switch
S3-S6	- Tactile switch
S7, S8	- On/off switch
	- 5×8-bit LED DIP array
	- Computer power supply (SMPS)
	- Wires and jumpers
	- 8×2-pin connector terminal

value for any random ROM address:
ROM2 ROM1
0b11111111 0b10110001

Challenges faced during project design and development

1. Designing the CPU architecture required logic implementation for modules and blocks
2. Programming the ROM for control logics during state change of a particular instruction
3. Fan In and Fan Out issues while driving logic signals in multiple chips
4. Noise and signal inductance issues in the circuit
5. Determining the minimum states

EFY Note

The source code and complete circuit diagram (Fig. 2) of this project are included in this month's

EFY DVD and are also available for free efymag.com

required for each instruction and modifying the architecture accordingly

6. Determining the sequence of operation for clearing different modules
7. Issues of common address/data bus architecture
8. Prototyping complex logic on the PCB/breadboard

Advantages of the project

1. Helps gain an understanding of the working of a microprocessor (all modules and instruction flow between modules)
2. Useful substitute for microprocessor trainer boards
3. Programming (feeding instructions and data) is easier
4. Clock speed can be adjusted to monitor instruction/data flow in the computer
5. Multiple displays provide the CPU information
6. In case of malfunction, IC replacement is cheaper and easier
7. A perfect kit for hobbyists and computer enthusiasts interested in retro programming/computing

Disadvantages

1. Instruction set limited to only 16
2. Displays are LED bar type, not 7-segment/readable
3. Input is through DIP switches, which is not very user-friendly
4. Program needs to be written every time the CPU goes off
5. Maximum result is 255 (8-bit only) **EFY**



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